

Lab 14: Guessing Who People Are

84-355 Democracy's Data | Week 14

Your name here

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Before we start

Voter files do not record race. In most states nobody is asked, and in the states that do ask, many people decline.

But the Voting Rights Act is about race. To argue in court that a districting plan dilutes Black or Latino voting strength, somebody has to estimate how many Black or Latino voters there are and how they voted — from records that do not say. The method the courts have largely accepted is the one you will build today: **Bayesian Improved Surname Geocoding**, from Imai and Khanna's 2016 paper, which you read for Tuesday.

It works by combining two things the government does publish: what a surname suggests, and where the person lives.

You are going to build it yourself, in three lines of arithmetic, and then spend the rest of the lab on the harder question of when it should be trusted.

Everything is base R. Work in order.

Part 1: What a surname knows

```
sn <- read.csv("data/census_surnames.csv", stringsAsFactors = FALSE)
dim(sn)
```

```
#> [1] 162254      9
```

Every surname held by **100 or more people** at the 2010 Census — 162,254 of them — with the racial and ethnic composition of the people who hold it.

```
head(sn, 3)
```

```
#>      name rank   count pctwhite pctblack pctapi pctaian pct2prace pthispanic
#> 1  SMITH    1 2442977   70.90   23.11   0.50   0.89     2.19     2.40
#> 2 JOHNSON    2 1932812   58.97   34.63   0.54   0.94     2.56     2.36
#> 3 WILLIAMS   3 1625252   45.75   47.68   0.46   0.82     2.81     2.49
```

The six columns are percentages of the people with that name, and they are mutually exclusive: non-Hispanic White, non-Hispanic Black, non-Hispanic Asian/Pacific Islander, non-Hispanic American Indian/Alaska Native, non-Hispanic two or more races, and Hispanic of any race.

Look one up:

```
look <- function(nm) sn[sn$name == toupper(nm), ]
```

```
look("NGUYEN")
```

```
#>      name rank   count pctwhite pctblack pctapi pctaian pct2prace pthispanic
#> 38 NGUYEN   38 437645    0.95    0.12  96.45    0.03     1.83     0.63
```

```
look("GARCIA")
```

```
#>      name rank   count pctwhite pctblack pctapi pctaian pct2prace pthispanic
#>  6 GARCIA    6 1166120    5.38    0.45   1.41    0.47     0.26    92.03
```

Ninety-six percent of Nguyens are Asian or Pacific Islander. Ninety-two percent of Garcias are Hispanic. For some names, the surname alone is nearly decisive.

Now try a few others:

```
rbind(look("SMITH"), look("JOHNSON"), look("WILLIAMS"))
```

```
#>      name rank   count pctwhite pctblack pctapi pctaian pct2prace pthispanic
#> 1  SMITH    1 2442977   70.90   23.11   0.50   0.89     2.19     2.40
#> 2 JOHNSON    2 1932812   58.97   34.63   0.54   0.94     2.56     2.36
#> 3 WILLIAMS   3 1625252   45.75   47.68   0.46   0.82     2.81     2.49
```

Williams is **45.8% white and 47.7% Black**. The name is a coin flip.

Williams is not an obscure case: it is the **third most common surname in the country**, held by 1.63 million people. So how much of America does the method fail on?

```
grp6 <- c("pctwhite", "pctblack", "pctapi", "pctaian", "pct2prace", "pcthispanic")
sn$top <- apply(sn[, grp6], 1, function(x) max(x, na.rm = TRUE))
ok <- is.finite(sn$top) & !is.na(sn$count)

cat("People whose surname is >90% one group:",
    format(sum(sn$count[ok & sn$top > 90]), big.mark = ","),
    sprintf("(%.1f%%)\n", 100 * sum(sn$count[ok & sn$top > 90]) / sum(sn$count[ok])))
```

```
#> People whose surname is >90% one group: 110,664,814 (37.5%)
```

```
cat("People whose surname never reaches 60%:",
    format(sum(sn$count[ok & sn$top < 60]), big.mark = ","),
    sprintf("(%.1f%%)\n", 100 * sum(sn$count[ok & sn$top < 60]) / sum(sn$count[ok])))
```

```
#> People whose surname never reaches 60%: 27,497,216 (9.3%)
```

For 37.5% of Americans a surname is nearly decisive. For 27.5 million — one in eleven — it settles nothing at all.

That is the shape of the whole method. BISG is not uniformly good or uniformly bad; it is excellent for Nguyen, useless for Williams, and the people it fails on are not a random sample of the country.

Try your own. Change the name below and re-run. Your name is in this file if at least a hundred Americans shared it in 2010.

```
look("CERVAS")
```

```
#>      name    rank count pctwhite pctblack pctapi pctaian pct2prace
#> 157113 CERVAS 156044   104    58.65      NA  34.62      0         NA
#>      pcthispanic top
#> 157113         5.77 58.65
```

Part 2: The asymmetry

How often does a surname actually settle the question? Count the names — restricting to those held by more than a thousand people, so we are not looking at noise — where one group makes up more than 90% of bearers.

```
big <- sn[sn$count > 1000, ]
nrow(big)
```

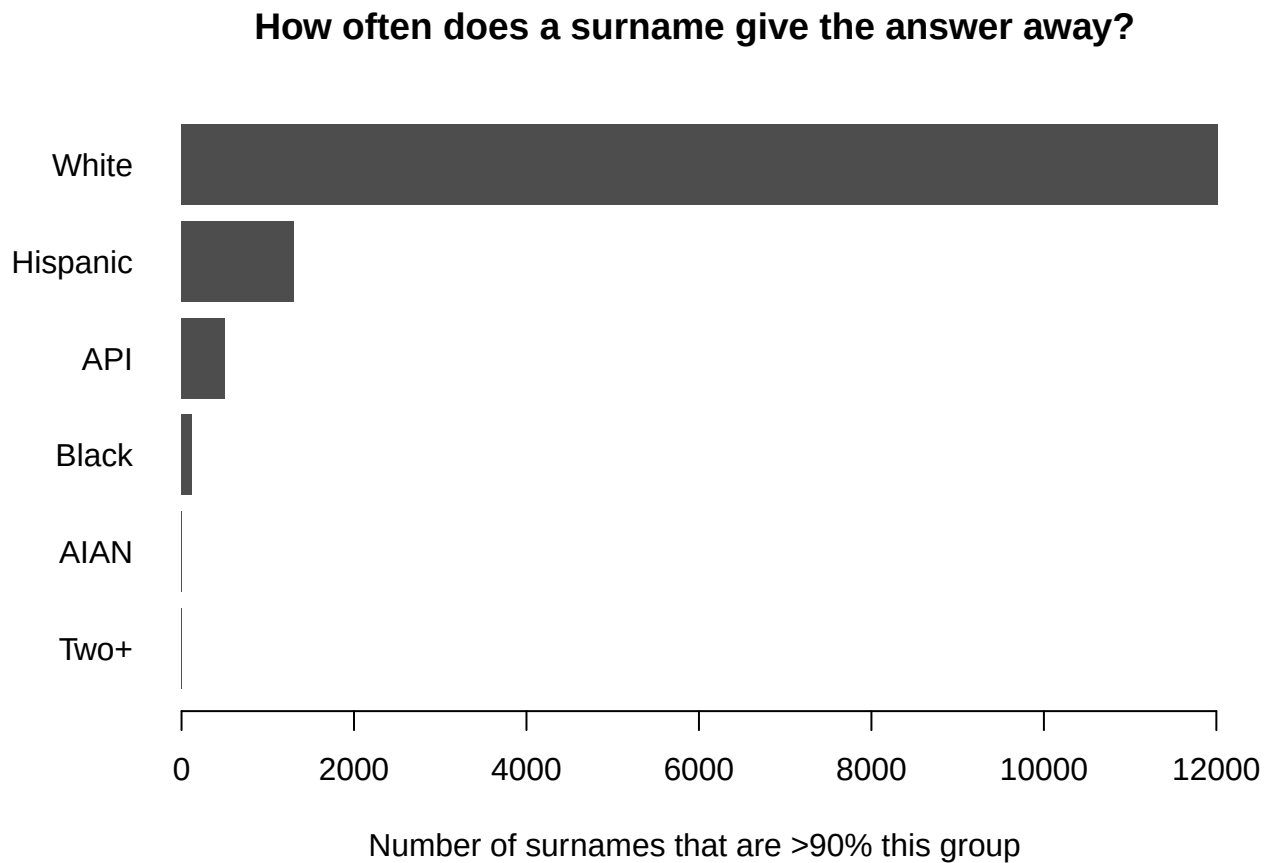
```
#> [1] 24874
```

```
grp <- c("pctwhite", "pctblack", "pctapi", "pctaian", "pct2prace", "pcthispanic")
sapply(grp, function(v) sum(big[[v]] > 90, na.rm = TRUE))
```

```
#>    pctwhite    pctblack    pctapi    pctaian    pct2prace pcthispanic
#>      12022        122        499          6          0        1303
```

```
n90 <- sapply(grp, function(v) sum(big[[v]] > 90, na.rm = TRUE))
names(n90) <- c("White", "Black", "API", "AIAN", "Two+", "Hispanic")
n90 <- sort(n90)
```

```
par(mar = c(5, 8, 4, 2))
barplot(n90, horiz = TRUE, las = 1, col = "#4D4D4D", border = NA,
        xlab = "Number of surnames that are >90% this group",
        main = "How often does a surname give the answer away?")
```



12,022 surnames are more than 90% white. Only 122 are more than 90% Black.

There are a hundred times more names that identify a white American than identify a Black American — and Black Americans are about an eighth of the country, not a hundredth.

This is not a quirk of the data. It is history sitting in a spreadsheet. Enslaved people were given, or took, the surnames of enslavers, and after emancipation those names stayed. Black and white Americans have shared a common stock of surnames for a hundred and sixty years.

The exceptions are worth seeing:

```
b <- big[!is.na(big$pctblack), ]
head(b[order(-b$pctblack), c("name", "count", "pctblack", "pctwhite")], 8)
```

```
#>      name count pctblack pctwhite
#> 16569  KOROMA  1733    97.75    0.75
#> 24110 ACHEAMPONG 1045    97.70    0.86
#> 24480  AGYEMANG 1023    97.46    0.78
#> 20524  WARSAME  1293    97.37     NA
#> 20969  MEKONNEN 1256    97.21    1.11
#> 23907  ANYANWU 1056    97.06    1.14
#> 23895  ALTIDOR  1057    96.78     NA
#> 10614  BOATENG  3017    96.75    1.36
```

```
rbind(look("WASHINGTON"), look("JEFFERSON"), look("LINCOLN"))
```

```
#>      name rank  count pctwhite pctblack pctapi pctaian pct2prace
#> 145  WASHINGTON 145 177386    5.17    87.53    0.30    0.68    3.78
#> 615  JEFFERSON  615  55179    17.45    74.24    0.40    1.90    3.54
#> 2209 LINCOLN 2209  16477    74.90    14.71    1.35    3.68    2.85
#>      pcthispanic top
#> 145          2.54 87.53
#> 615          2.47 74.24
#> 2209          2.51 74.90
```

Washington is the most identifiably Black common surname in America — 87.5% of the 177,386 people who carry it. Jefferson is 74.2%.

These names were chosen. After emancipation many freed people took the names of presidents associated with liberty, and the pattern is still legible in the 2010 Census.

Notice that Lincoln is only 14.7% Black — far below the other two. The adoption was not uniform, and nobody planned it. Whatever produced this pattern was millions of separate decisions, and it is still measurable a century and a half later.

Part 3: Where you live

The second ingredient.

```
co <- read.csv("data/county_race.csv", stringsAsFactors = FALSE,
              colClasses = c(fips = "character"))
dim(co)
```

```
#> [1] 3144  10
```

```
head(co[, c("fips", "county", "state", "pop", "white", "black", "hispanic")], 3)
```

```
#>   fips      county      state   pop   white   black  hispanic
#> 1 01001 Autauga County Alabama 61464 0.7093421 0.21277496 0.04010478
#> 2 01003 Baldwin County Alabama 261608 0.8174903 0.08310908 0.06155393
#> 3 01005 Barbour County Alabama 24358 0.4486411 0.46145004 0.06474259
```

Every county, in the same six categories. Two examples:

```
co[co$fips %in% c("42003", "36005"),
  c("county", "state", "pop", "white", "black", "api", "hispanic")]
```

```
#>           county      state    pop    white    black    api
#> 1833    Bronx County    New York 1384724 0.08874404 0.2962265 0.04668656
#> 2248 Allegheny County Pennsylvania 1231814 0.75992642 0.1310928 0.04955943
#>           hispanic
#> 1833 0.55424547
#> 2248 0.03277199
```

Allegheny County is 76% white and 13% Black. The Bronx is 9% white, 30% Black, and 55% Hispanic. A name means something different in each.

Part 4: Bayes' rule, by hand

Here is the whole method.

We want $P(\text{race} \mid \text{surname}, \text{county})$. We have $P(\text{race} \mid \text{surname})$ from the surname file and $P(\text{race} \mid \text{county})$ from the county file. Bayes' rule combines them:

$$P(r \mid s, g) \propto \frac{P(r \mid s) P(r \mid g)}{P(r)}$$

where $P(r)$ is the national share of group r . Dividing by it prevents double-counting the fact that some groups are large everywhere.

```
grp6 <- c("white", "black", "api", "aian", "twoplus", "hispanic")
natl <- colSums(co[, grp6] * co$pop) / sum(co$pop)
round(natl, 4)
```

```
#>   white   black    api    aian twoplus hispanic
#> 0.5746 0.1263 0.0669 0.0072 0.0248 0.2002
```

That is the entire model:

```
bisg <- function(name, fips) {
  s <- sn[sn$name == toupper(name), ]
  p_s <- c(s$pctwhite, s$pctblack, s$pctapi,
    s$pctaian, s$pct2prace, s$pcthispanic) / 100
  p_s[is.na(p_s)] <- 0 # suppressed cells contribute nothing

  p_g <- as.numeric(co[co$fips == fips, grp6])

  u <- p_s * p_g / natl # <- Bayes' rule, one line
  setNames(round(100 * u / sum(u), 1), grp6)
}
```

Three lines. Now watch what it does.

```
bisg("SMITH", "42003")      # Allegheny County, PA
```

```
#>   white   black   api   aian twoplus hispanic  
#>   77.5   19.8   0.3   0.1    1.9    0.3
```

```
bisg("SMITH", "36005")      # Bronx County, NY
```

```
#>   white   black   api   aian twoplus hispanic  
#>   14.9   73.7   0.5   0.5    1.3    9.0
```

Smith is 70.9% white nationally. A Smith in the Bronx is 73.7% Black.

The name did not change. What changed is that we stopped ignoring where the person lives.

```
bisg("LEE", "42003")       # Allegheny County, PA
```

```
#>   white   black   api   aian twoplus hispanic  
#>   48.1   17.1  31.6   0.2    2.7    0.3
```

```
bisg("LEE", "15003")       # Honolulu County, HI
```

```
#>   white   black   api   aian twoplus hispanic  
#>    3.0    0.9  90.0   0.1    5.7    0.3
```

Lee is 42% Asian and 36% white by name alone. In Allegheny County it resolves to white; in Honolulu it resolves to 90% Asian. The same four letters support opposite conclusions.

Part 5: When it fails

A third of the surname file is missing.

```
cells <- nrow(sn) * length(grp)  
miss  <- sum(is.na(sn[, grp]))  
c(cells = cells, suppressed = miss, pct = round(100 * miss / cells, 1))
```

```
#>   cells suppressed   pct  
#> 973524.0  315341.0  32.4
```

The Census suppresses any cell small enough to risk identifying someone. Those appear as NA, and in the function above they contribute zero — which quietly assumes the true value is zero when what we actually know is *small and unknown*. For rare names this matters.

```
bucket <- cut(sn$count, breaks = c(0, 200, 500, 1000, 5000, 1e7),  
              labels = c("<200", "200-500", "500-1k", "1k-5k", ">5k"))  
supp <- 100 * tapply(is.na(sn$pctblack), bucket, mean)  
round(supp, 1)
```

```
#>    <200 200-500 500-1k 1k-5k    >5k
#>    42.1    49.1    53.3    28.0    1.7
```

For surnames held by fewer than two hundred people, the Black percentage is suppressed 42.1% of the time. **That is the headline number, and it is not the worst one.** Suppression *rises* to 49.1% and peaks at 53.3% in the 500-1k bucket, then falls away to 28.0% and 1.7%. **The method is not simply least informative for the rarest names** — the loss of information is not a smooth decline at all. And rarity is not evenly distributed across groups.

And a probability is not a person. The output of `bisg()` is a statement about a population: *of all the Smiths in the Bronx, roughly 74% are Black*. It is not a statement about any individual Smith. Every use of this method involves taking a number that is true on average and attaching it to somebody in particular.

That matters because of what it is used for. BISG is used to estimate racial voting patterns in Voting Rights Act litigation — the evidence on which districts are drawn or struck down. It is used by the Consumer Financial Protection Bureau to detect lending discrimination. In both cases a real decision about real people rests on an inference from a name and a zip code.

The alternative is not a better measurement. The alternative is **no measurement**, and therefore no ability to detect discrimination at all. That is the trade Imai and Khanna's paper is really about, and it is worth being clear that it is a trade rather than a solution.

Part 6: On your own

This is the part you turn in. **You are not being asked to write code.** Each option is a working chunk with one thing to change.

Pick **one**.

Option A — One name, many places

```
my_name <- "WILLIAMS"      # <- change this. Try your own name.

places <- c("42003", "36005", "15003", "48141", "06037", "36061")
labels <- c("Allegheny PA", "Bronx NY", "Honolulu HI",
            "El Paso TX", "Los Angeles CA", "Manhattan NY")

out <- t(sapply(places, function(f) bisg(my_name, f)))
rownames(out) <- labels
out
```

```
#>           white black api aian twoplus hispanic
#> Allegheny PA  53.2  43.5 0.3  0.1     2.5      0.4
#> Bronx NY      5.5  87.6 0.3  0.3     1.0      5.4
#> Honolulu HI   27.7  18.9 7.0  0.4    43.5      2.4
#> El Paso TX    25.7  39.7 0.3  1.0     3.1     30.3
#> Los Angeles CA 33.5  49.6 1.8  0.4     4.6     10.2
#> Manhattan NY  39.7  52.9 1.0  0.2     2.9      3.3
```

```
look(my_name)
```

```
#>      name rank   count pctwhite pctblack pctapi pctaian pct2prace pcthispanic
#> 3 WILLIAMS    3 1625252   45.75   47.68   0.46    0.82     2.81       2.49
#>      top
#> 3 47.68
```

Write about: how much does the estimate move across these six counties? Which county changes it most, and why? Then the question that matters: if this name were on a voter file and a court needed to know the voter's race, how much of the answer would come from the name and how much from the address? Would you be comfortable with the result being used as evidence?

Option B — Which names does geography rescue?

```
test_names <- c("SMITH", "WILLIAMS", "JOHNSON", "LEE",
               "NGUYEN", "GARCIA", "WASHINGTON") # <- edit this list

fips_here <- "36005" # <- and this county

for (nm in test_names) {
  s <- look(nm)
  before <- max(c(s$pctwhite, s$pctblack, s$pctapi, s$pcthispanic), na.rm = TRUE)
  after <- max(bisg(nm, fips_here))
  cat(sprintf("%-12s strongest guess: %5.1f%% -> %5.1f%% (%+.1f)\n",
              nm, before, after, after - before))
}
```

```
#> SMITH          strongest guess:  70.9% ->  73.7%  (+2.8)
#> WILLIAMS       strongest guess:  47.7% ->  87.6% (+39.9)
#> JOHNSON        strongest guess:  59.0% ->  82.2% (+23.2)
#> LEE            strongest guess:  42.2% ->  47.8% (+5.6)
#> NGUYEN         strongest guess:  96.5% ->  95.7% (-0.8)
#> GARCIA         strongest guess:  92.0% ->  98.8% (+6.8)
#> WASHINGTON     strongest guess:  87.5% ->  95.4% (+7.9)
```

```
co[co$fips == fips_here, c("county", "state", "white", "black", "api", "hispanic")]
```

```
#>      county  state  white  black  api  hispanic
#> 1833 Bronx County New York 0.08874404 0.2962265 0.04668656 0.5542455
```

Write about: for which names does adding geography sharpen the estimate, and for which does it barely help? You should find that names which were already decisive (Nguyen, Garcia) gain little, while ambiguous ones (Smith, Williams) gain a great deal. Explain why that is exactly what Bayes' rule should do. Then say which kind of name is more common in America, and what that implies about how much of BISG's accuracy is really coming from the address rather than the name.

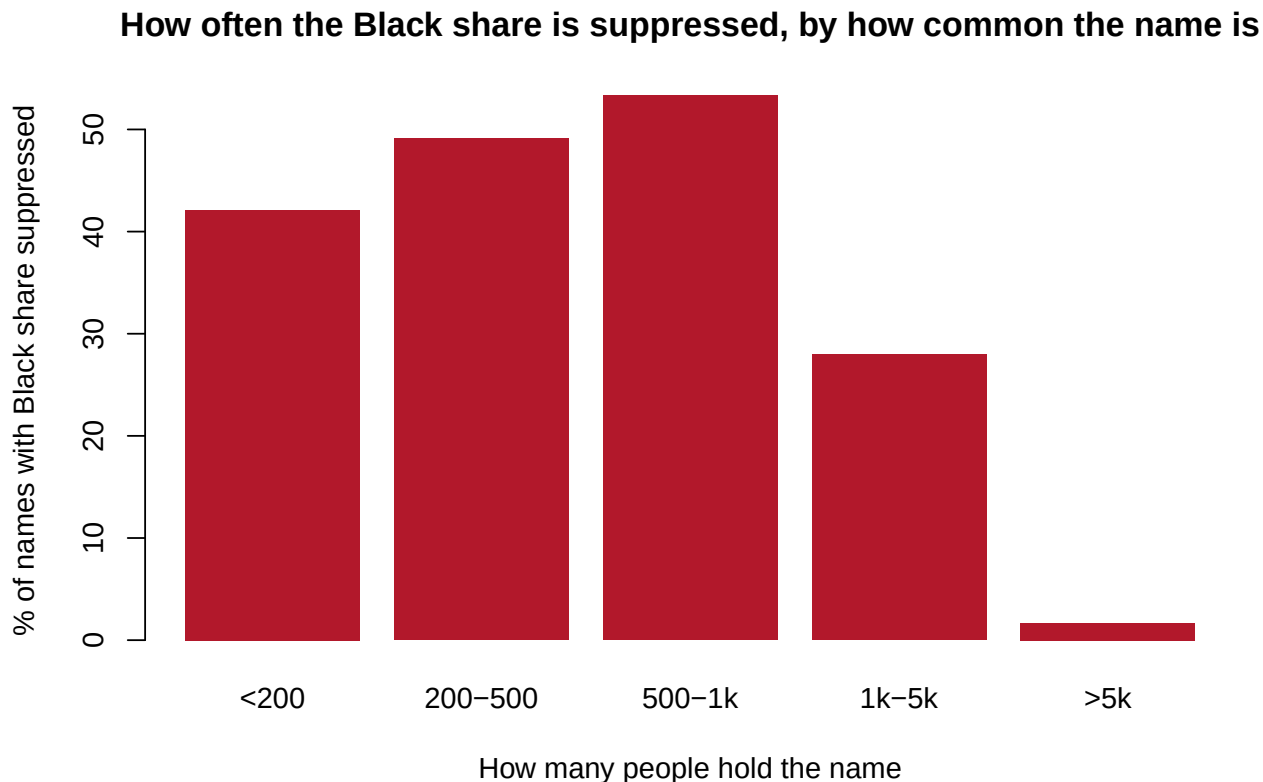
Option C — The cost of suppression

```
cutoff <- 1000          # <- change this

sn$bucket <- cut(sn$count, breaks = c(0, 200, 500, 1000, 5000, 1e7),
                 labels = c("<200", "200-500", "500-1k", "1k-5k", ">5k"))
supp <- tapply(is.na(sn$pctblack), sn$bucket, mean)
round(100 * supp, 1)
```

```
#>    <200 200-500 500-1k 1k-5k    >5k
#>   42.1   49.1   53.3   28.0    1.7
```

```
barplot(100 * supp, col = "#B2182B", border = NA,
        ylab = "% of names with Black share suppressed",
        xlab = "How many people hold the name",
        main = "How often the Black share is suppressed, by how common the name is")
```



```
c(names_below_cutoff = sum(sn$count < cutoff),
  share_of_all_names = round(100 * mean(sn$count < cutoff), 1))
```

```
#> names_below_cutoff share_of_all_names
#>          137364.0           84.7
```

Write about: describe the pattern, and read the printed numbers before you do — suppression does not simply fall away as names get more common, and the rarest bucket is not the worst one. Now think about who has a rare surname in the United States — recent immigrants, small national-origin groups, Native

American families. Is a method whose precision depends this much on how common your name is neutral with respect to who it works worst *for*? Say what you would want a court to be told about this before the estimate was admitted.

Option D — Ask your own question

Look up your family's names, or the names of people in the news. Find a name whose estimate you think is wrong and work out why. Compare two counties you know.

What to submit. Knit to HTML or PDF and upload to Canvas, with your chosen chunk and **two or three paragraphs** of interpretation.

The writing is the assignment. I am not grading your code — it is my code. I am grading whether you can hold two things at once: that this method is genuinely useful, and that it is genuinely a guess.

Where the data came from

Surnames — U.S. Census Bureau, *Frequently Occurring Surnames from the 2010 Census* (<https://www2.census.gov/topics/genealogy/2010surnames/names.zip>). Every surname held by 100 or more people, with the racial composition of bearers. Cells small enough to risk identifying an individual are suppressed.

County composition — U.S. Census Bureau, Population Estimates Program, county characteristics file for 2024, collapsed to the same six categories.

The method — Kosuke Imai & Kabir Khanna, “Improving Ecological Inference by Predicting Individual Ethnicity from Voter Registration Records,” *Political Analysis* 24 (2016): 263–72, which you read for Tuesday.

Two notes on what we did *not* use. The published `wru` package implements this method properly, including first names, age, sex, and party registration, and geography down to the census block. We have implemented the county-level core by hand instead, so that the arithmetic is visible rather than hidden inside a function call — and so that this lab needs no packages and no API key.

The real thing is better than what you built today. It is better in degree, not in kind: it is still a guess about a person, assembled from facts about populations.

The script that downloaded, checked and wrote these files is `data/build-data.R`.